

Imperial College London

BSc/MSci EXAMINATION June 2025

This paper is also taken for the relevant Examination for the Associateship

STATISTICAL METHODS FOR EXPERIMENTAL PHYSICS

For MLBD MRes Students

June 4th, 2025: 10:00 to 12:00

Answer all FOUR questions

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 4 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 4 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. The exponential decay of a radioactive isotope is described by the exponential probability density function (PDF) given by

$$f(t) = \lambda e^{-\lambda t} \quad (t \geq 0),$$

where t is the time of decay of the isotope and $\lambda > 0$ is the decay constant.

- (i) Show that the expectation value of the decay time t is given by

$$E[t] = \frac{1}{\lambda}.$$

HINT: You may find the following integral useful:

$$\int_0^{\infty} x e^{-\alpha x} dx = \frac{1}{\alpha^2} \quad (\alpha > 0).$$

[2 marks]

- (ii) Using the method of moments, find an estimator of the decay constant λ from a set of N independent measurements of the decay time t_i , where $i = 1, 2, \dots, N$.

[3 marks]

- (iii) If t_1 and t_2 are the decay times of two isotopes with the same decay constant λ , show that the probability of observing $T = t_1 + t_2$ is given by

$$f(T) = \lambda^2 T e^{-\lambda T} \quad (T \geq 0)$$

if the decay times are independent.

[5 marks]

Find the maximum likelihood estimate of the decay constant λ from a set of N independent measurements of the sum of decay times T_i , where $i = 1, 2, \dots, N$.

[4 marks]

- (iv) Show that the maximum likelihood estimate of the decay constant λ that you derived in the previous question is a biased estimator. HINT: You may find the following inequality useful:

$$E \left[\frac{1}{\sum x} \right] \geq \frac{1}{E[\sum x]}.$$

[8 marks]

- (v) Typically, *good* estimators are unbiased. Name two other properties of good estimators, and describe what they mean.

[3 marks]

[Total 25 marks]

2. The Poisson probability distribution for a discrete random variable k is given by

$$p(k; \lambda) = e^{-\lambda} \frac{\lambda^k}{k!}$$

- (i) Show that the maximum likelihood estimator for the parameter λ , assuming an observed value of k is given by

$$\hat{\lambda} = k.$$

[3 marks]

- (ii) A series of n independent sequential measurements is made of the number of light pulses from a distant galactic source in a fixed interval of time. Each measurement is an integer number k_i , for $i = 1 \dots n$, which is distributed as a random Poisson variable with parameter λ_i . Figure 1 shows a set of such measurements. It is assumed that the source pulses at a constant rate of T pulses over a single measurement interval of time. Under this assumption $\lambda_i = T$ for every i .

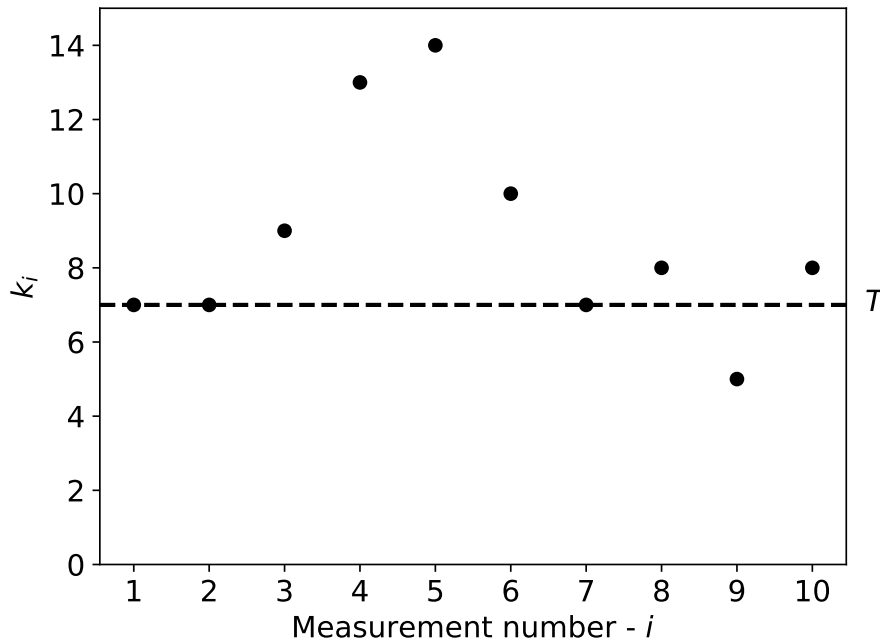


Figure 1

- (a) Write down the likelihood function for the set of n measurements as a function of T , assuming a constant rate T .
- (b) Find the expression for the maximum likelihood estimator $\hat{T}$ given the measurements.
- (c) Describe how you would determine a 95% confidence region for the parameter T from the measurements using the Neyman construction and the profile likelihood ratio test statistic,

$$t = -2 \ln \left(\frac{L(T)}{L(\hat{T})} \right).$$

[This question continues on the next page ...]

[11 marks]

- (iii) An alternate hypothesis for the measured pulses is that the rate of pulses of the source varies with time. Figure 2 shows the same measurements but with the two hypotheses overlaid: H_0 represents the constant rate hypothesis and H_1 represents an alternative hypothesis assuming a time varied pulse rate.

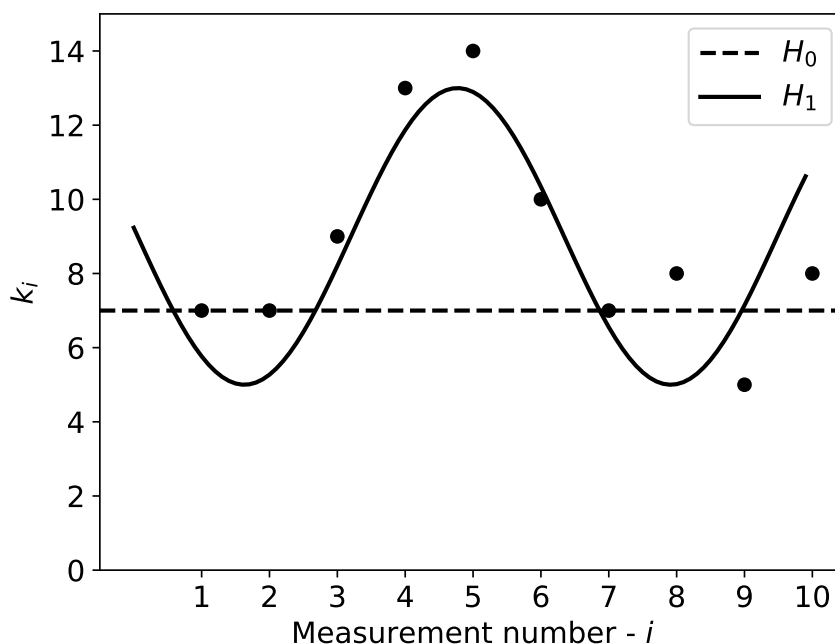


Figure 2

- (a) State the definition of Type-I and Type-II errors in terms of the two hypotheses H_0 and H_1 .
- (b) Devise a hypothesis test to test if H_0 is a good description of the data, based on a likelihood ratio test statistic,

$$t = -2 \ln \left(\frac{L(H_1)}{L(H_0)} \right)$$

and describe how you would perform the test using the data shown in Figure 2. You should *clearly* describe each step of the test, explaining how to generate distributions of your test statistic, and define your type-I error and how you would calculate the power of your test. You do not need to calculate or write out in full the likelihoods or the test statistic. You can use diagrams to help describe your procedures.

[11 marks]

[Total 25 marks]

3. (i) (a) What is meant by a *conjugate prior*?
 (b) What is the technical distinction between $p(\text{data}|\text{parameters})$ when it is treated as a sampling distribution and as a likelihood? [6 marks]
- (ii) A polling organisation wishes to predict the outcome of the next election, which is being contested by two parties, A and B . Its goal is to infer the probability q that electors will vote for party A . It polls N people, who answer truthfully and independently of each other who they would vote for. n people respond that they would vote for party A .

Argue that the probability of this happening is the binomial distribution

$$p(n) = \binom{N}{n} q^n (1-q)^{N-n}.$$

[5 marks]

- (iii) (a) In the absence of any other information, what would be a suitable prior for q ? Normalise it correctly.
 (b) The polling organisation has some prior information from previous surveys, and chooses a Dirichlet prior $\pi(q) = D(q; \alpha, \beta)$, which is defined by

$$\pi(q) = D(q; \alpha, \beta) \propto q^\alpha (1-q)^\beta,$$

and α and β are (hyper)parameters of the prior. The constant of proportionality ensures that the prior is normalised, but is not relevant and can be ignored.

What values of α and β correspond to a uniform distribution?

- (c) Show that the posterior is given by another Dirichlet distribution, and find expressions for the posterior parameters in terms of α, β, n and N . [7 marks]
- (iv) (a) Show that the maximum a posteriori (MAP) value of q in this case is given by

$$q_{\text{MAP}} = \frac{\alpha + n}{\alpha + \beta + N}.$$

- (b) By considering the second derivative of the log posterior, find an expression for the approximate posterior variance of q .

[7 marks]

[Total 25 marks]

4. (i) (a) From the laws of probability, derive Bayes theorem

$$p(X|Y) = \frac{p(Y|X) p(X)}{p(Y)}.$$

- (b) A faulty theft detection device at the exit of a department store incorrectly goes off when an innocent shopper leaves 2% of the time. It will also fail to detect a real shoplifter in 10% of cases. If 99% of shoppers are honest, use Bayes theorem to calculate the probability that a shopper is a shoplifter if the alarm goes off. [8 marks]
- (ii) (a) When comparing two models M_0 and M_1 , the posterior odds for each model, $p(M_i|d)$ (given some data d) may be calculated using Bayes theorem. Show that the relative posterior odds is equal to the Bayes factor (the ratio of the Bayesian evidence values $p(d|M_i)$) if a condition is satisfied (you should state this), and derive an expression for the Bayesian evidence involving its model parameters.
- (b) What requirement is placed on the prior on the parameters in order to compute the Bayesian evidence? [8 marks]
- (iii) (a) A man arrives in a city and takes a random taxi with city licence number n to his hotel. Assuming that licences are issued in sequence from 1 upwards, what is the posterior probability for the total number of taxis in the city, N ? Assume a uniform prior for N (truncated at some large value), and there is no need to normalise the posterior (or the prior).
- (b) What is the maximum likelihood estimate for N ?
- (c) Comment on the merits of the Bayesian approach to a frequentist maximum likelihood estimator for this problem.

[9 marks]

[Total 25 marks]